

STEP 1: ANALYSE PROBLEM STATEMENT

1.1 INTRODUCTION

You have been provided with a generic, simulated dataset "Sales Case Study.xlsx", which indicates the daily trading information for a large retail store, aggregated for the day. The data is only for one of the products sold at this store.

Each record contains the:

- Date = the day on which the sales occurred
- Sales = total Rand value of the sales that occurred
- Cost of Sales = total Rand value of the cost of sales that occurred
- Quantity Sold = total number of units that have been sold

1.2 INSTRUCTIONS AND OBJECTIVE

Using this dataset, your objective is to develop metrics and derive subsequent insights. The metrics that you need to develop are:

1. What is the daily sales price per unit?
2. What is the average unit sales price of this product?
3. What is the daily % gross profit?
4. What is the daily % gross profit per unit?
5. Pick any 3 periods during which this product was on promotion/special
- a. What was the Price Elasticity of Demand during each of these periods?
- b. In your opinion, does this product perform better or worse when sold at a promotional price?
6. Please derive any other interesting insight you can from the dataset provided. This can include:
 - a. Interesting Visuals
 - b. Reports
 - c. Dashboards
 - d. KPIs or metrics.Important to note:
 - You will have 2 working days, from the day that you receive this case study, to submit your work.
 - It is important to present/submit your work in a readable and understandable manner, as you will be assessed on not only the answers provided but also the process followed.
 - You may use any tool at your disposal, and you are also allowed to research any techniques required to develop your work.

Step 2: Ask data-driven questions

Promotions & elasticity

- Which periods show a sustained drop in price that looks like a promotion, rather than daily noise?
- When price drops during those periods, how much does quantity sold increase?
- Is demand elastic or inelastic during those periods — does a small price cut produce a proportionally larger increase in volume?
- Does total gross profit (in Rand) go up or down during a promo, even though the profit % per unit falls?

Pricing & unit economics

- What is the daily price per unit, and how much does it vary day to day?
- What is the Rand-weighted average unit price across the full 3 years, and how does that differ from a simple average of daily prices?
- Is the price trending up, down, or flat over the 3-year period?

Volume & seasonality

- Are there day-of-week patterns in how much is sold?
- Are there month-of-year or seasonal patterns (e.g. higher volume around certain months each year)?
- Are there individual days that look like outliers (extreme spikes or drops) worth investigating separately?

Profitability

- What is the daily % gross profit, and how does it vary over time?
- Is the product profitable overall — does total Sales exceed total Cost of Sales across the whole dataset?
- Are there specific months or periods where profitability turns negative, and do they cluster around anything (season, price level, promo activity)?

This product should not be promoted further until its underlying cost structure is fixed.

Step 3: Analysis Planning

1. Pricing & unit economics

- *Method:* Direct formula, daily grain
— $\text{price_per_unit} = \text{Sales} / \text{Quantity Sold}$. Average = $\text{SUM}(\text{Sales}) / \text{SUM}(\text{Quantity})$, not $\text{AVG}(\text{daily price})$.
- *Trend check:* Simple line chart of price over time; optionally a rolling 14/30-day average to smooth noise.
- *Tool:* SQL view (already built) → line chart in SQL Editor or Excel.

2. Profitability

- *Method:* $\text{gross_profit} = \text{Sales} - \text{Cost of Sales}$; $\text{gross_profit_pct} = \text{gross_profit} / \text{Sales}$. Aggregate to monthly for the trend view (daily is too noisy to read).
- *Overall check:* single SUM comparison — total Sales vs total Cost of Sales.
- *Tool:* SQL (monthly GROUP BY) → bar/line combo chart; mirrored in Excel Monthly Pivot.

3. Promotions & elasticity

- *Method:* Detect promo windows statistically (price meaningfully below a baseline — either global mean or rolling mean), then cluster consecutive flagged days into periods.
- *Elasticity:* For each chosen period, compare average price/quantity in the promo window to a baseline window (e.g. 14 days immediately prior). $\text{PED} = \% \Delta \text{Quantity} / \% \Delta \text{Price}$.
- *Validation step:* Cross-check the picks with a second method (you already have this — rolling z-score in PySpark) rather than trusting one detection pass.
- *Tool:* SQL views → bar charts comparing baseline vs promo.

4. Volume & seasonality

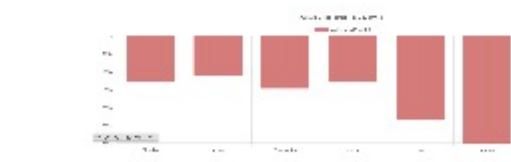
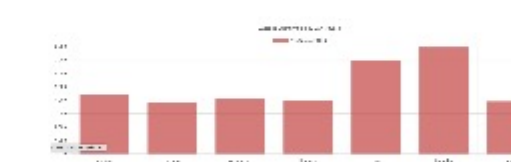
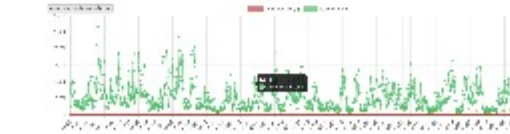
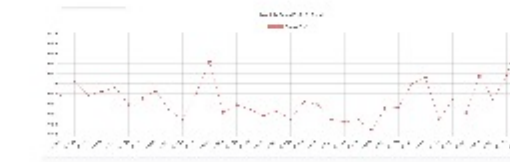
- *Method:* GROUP BY day-of-week and GROUP BY month; compare average quantity/price/GP across groups.
- *Outliers:* Sort by gross_profit_pct or quantity ascending/descending, inspect the extremes manually.
- *Tool:* SQL GROUP BY → bar charts; Excel pivot-style tables as backup/cross-check.

Step 4: Data Exploration

To be completed in Excel and Databricks- Exploring relationships in dataset

Avg unit price: **R35.40** (Rand-weighted: total Sales ÷ total Quantity Sold)
Overall gross profit %: **-3.8%**
Profitable overall! **N** — total Cost of Sales (R194.0m) exceeds total Sales (R186.9m); total gross loss = R7.1m across all 1,053 days
3 promo periods used: **24 Sep-6 Oct 2014 / 23 Sep-6 Oct 2015 / 23 Jun-6 Jul 2016**
PED for each: **-46.2 / -34.4 / -11.3**
Elastic or inelastic? **Elastic** — |PED| well above 1 in all three periods; small price cuts produced much larger jumps in quantity sold
Best/worst day of week: **Best (highest avg volume) = Saturday** (~7,937 units/day); **Worst (lowest avg volume) = Tuesday** (~3,852 units/day). Friday is a close second best.
One surprising insight: **the product is losing money overall despite a "normal-looking" average price** — 24 of 36 months ran at negative gross profit %, and the loss is concentrated on the highest-volume days (Fri/Sat), meaning the busiest trading days are also the biggest drag on profit rather than the strongest performers.

Step 5: Data Visualisation



Step 6: Feature Engineering

1 Core unit economics — price_per_unit, cost_per_unit, gross_profit, gross_profit_pct (already built earlier)
2 Calendar features — day_of_week, is_weekend, month, month_num, year
3 Rolling/trend features — 14-day rolling average and std dev of price/quantity, used as a "local normal" to compare each day against
4 Day-over-day change — price_change_pct, qty_change_pct (spot sudden jumps)
5 The promo flag — the key engineered feature: a 0/1 flag marking days priced meaningfully below their own rolling average
6 Combined view — vw_sales_features_full, one view bundling all of the above so later steps don't repeat the logic

Step 7: Communication Findings

sales_case_study_analysis.xlsx — daily metrics, promo/elasticity analysis, monthly and day-of-week pivots, charts
sales_case_study_analysis_databricks_matched.sql — full SQL matched to your actual table/columns
sales_exploration_pyspark.py — Databricks notebook with rolling averages, promo detection, and the elasticity cross-check
sales_case_study_deck.pptx — 10-slide presentation, headline-first, with the recommendation at the end